

Limit Theorems for Random Variables:
Theory and Simulation
MA300

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March 27, 2026

Abstract

This paper demonstrates the computational and visual verification of probability theorems using Python simulations and thorough mathematical proofs. This is achieved by modelling the behaviour of sample means across vast numbers of independent and identically distributed (i.i.d.) random variables. By processing large quantities of data, we graphically prove the mechanics of how both the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT) behave, further illustrating concepts such as *almost sure convergence* and *convergence in probability*.

Prior to pushing these theorems to their mathematical limits, this project contextualises their theoretical mechanics through tangible, real world applications. By examining how these probabilistic models govern everyday phenomena, ranging from demographic surveying to machine learning algorithms, we establish an intuitive foundation for the reader. This approach bridges the gap between abstract mathematical notation and observable reality, ensuring that the strict mathematical proofs are grounded in practical utility before they are subjected to computational stress testing.

Additionally, instead of limiting the scope to distributions where these theorems succeed, this project investigates their mathematical boundaries. Throughout this research we show that visualising when these theorems fail provides a deeper understanding of statistics rather than observing their success alone. We explore this by introducing heavy-tailed distributions, specifically Cauchy and Pareto, into our testing environment. This allows us to observe the exact moments when these theorems mathematically collapse, as a result of infinite expected values and infinite variances, highlighting the imperfections of attempting to apply the limit theorems to highly volatile data.

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1 Introduction

Probability theory and statistics has a number of real world applications such as demographic surveying, national polling, manufacturing components, casinos and machine learning algorithms, which we cover in more depth in Section 2.3. At the heart of these predictive models lie the fundamental limit theorems, which serve as mathematical foundations for statistics. In practice, these theorems are often incorrectly applied to datasets without fully understanding the underlying boundaries required for them to function.

The primary motivation for this project stems from the widespread reliance on limit theorems and great excitement to translate this mathematics into a more tangible and visually understandable format. Simulations provide a unique perspective to explore these concepts, providing an alternative way to observe the behaviour of random variables on an enormous scale that would otherwise be impossible to calculate by hand.

This project focuses on investigating two central pillars of probability theory: the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT). To provide a comprehensive understanding of these theorems, this project evaluates them through two approaches. The first approach relies on rigorous mathematical proofs to establish the theoretical guarantees of convergence. The second utilises Python based simulations to graphically illustrate these theorems in action. Alongside proving the success of these theorems under standard conditions, this study will intentionally disrupt the ideal environment to observe the exact mathematical thresholds where the LLN and CLT fail.

Furthermore, the transition from continuous theoretical mathematics to discrete computational environments presents its own unique challenges. To accurately model these theorems, this research leverages simulation techniques and random number generation to process extensive synthetic datasets. By systematically comparing standard normal variables against Cauchy and Pareto distributions, we explicitly highlight the computational consequences of undefined expected values and infinite variances. This methodology not only verifies the asymptotic behaviour dictated by the limit theorems, but also provides a dynamic environment to safely observe statistical anomalies that would otherwise compromise real world predictive models.

2 The Law of Large Numbers

2.1 Historical development and background

For centuries, the LLN was understood only empirically, it remained merely an intuitive observation among gamblers [1, p.410-413]. The journey from this intuition has led to rigorous mathematical theorems which have bifurcated the theorem into two distinct forms: the Weak Law and the Strong Law.

The LLN has altered significantly over time, with multiple breakthroughs changing how we understand the theorem today. An example in which this can be seen is in the pre-

calculus era, where mathematicians such as Gerolamo Cardano (1501-1576) wrote about the regularity of dice rolls in his book "Games of Chance" [1, p.417]. In his book, he recognised that certain outcomes were more expected than others over many trials; at this stage, this notion was only empirical.

It wasn't until 1713 where the first rigorous step was taken by Jacob Bernoulli, where he provided the proof for what is now recognised as the Weak Law of Large Numbers (WLLN) for binary variables [6]. Bernoulli proved that for any small error margin, the probability that the sample proportion deviates from the true probability tends to zero as the sample size increases, this established Convergence in Probability [10]. It provided the first mathematical guarantee that a sufficiently large sample size renders the probability of error arbitrarily small. This justified the use of statistical sampling, proving that observing a finite number of trials could yield reliable information about an unknown probability, bridging the gap between empirical observation and theoretical probability [10].

In 1867, Chebyshev generalised Bernoulli's work using what is known today as Chebyshev's inequality [3]. Unlike Bernoulli's approach, Chebyshev's method relied only on using the existence of the mean and variance. This proof made the WLLN applicable to a count-less range of random variables beyond only simple coin flips, solidifying it as a fundamental tool in probability theory [3].

Whilst the WLLN was successful, it only guaranteed that the probability of a large error becomes negligible at a specific moment in time, it did not guarantee that the succession of averages would eventually stop fluctuating. In 1909, Émile Borel filled this gap by formulating the Strong Law of large numbers (SLLN), for specifically coin tossing, proving that the sample mean converges to the true mean with probability one *almost surely* [8]. This work was consolidated in the 1930's by Andrey Kolmogorov, who established the necessary conditions for the SLLN to hold [8].

2.2 Weak and Strong Law (Conceptual Understanding)

To be able to grasp a well rounded understanding for this section, we will need to clearly distinguish between the WLLN and SLLN, first we need to outline their respective theoretical definitions and establish the mathematical notation used throughout this paper.

2.2.1 Foundational Theorems

Definition 2.1. For the following theorems, let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with finite expected value $\mathbb{E}[X_i] = \mu$ and finite variance $\text{Var}(X_i) = \sigma^2$, as stated in [7, sec.22].

Furthermore, let the sample mean be expressed by:

$$(2.1) \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Theorem 2.2 (WLLN). *Under the conditions defined above, the WLLN states that for any margin of error $\epsilon > 0$, the sample mean converges to the true mean μ as $n \rightarrow \infty$. This can be seen from:*

$$(2.2) \quad \lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = 0$$

Theorem 2.3 (SLLN). *Under the conditions defined above, the SLLN states that the sample mean converges almost surely to the true mean μ . [7] Mathematically, this is expressed as:*

$$(2.3) \quad \mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

With this foundation we can now explore the ideas of *convergence in probability* and *almost sure convergence* in the following sections.

2.2.2 Convergence in Probability

This is a concept centred around the WLLN. Fundamentally, it describes the reliability of a sample mean as the size of the sample grows. To understand its mechanics, we relate directly back to the limit equation established in Theorem 2.2:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = 0$$

This equation systematically breaks down the concept of statistical error. As the sample size (n) approaches infinity, the probability ($\mathbb{P}$) of a significant statistical error shrinks to exactly zero. Specifically, for any arbitrarily small margin of error ($\epsilon > 0$), the chance that our observed sample mean ($\bar{X}_n$) deviates from the true population mean (μ) by more than this margin ends up being negligibly rare. This guarantees that as datasets grow larger, estimates that deviate significantly from the true mean become highly improbable.

However, the crucial theoretical nuance of *convergence in probability* is that it evaluates the data at a specific snapshot in time. It provides a highly reliable probability for a single strictly defined sample size n . It does not evaluate the sequence continuously, so if the sequence were to continue infinitely, the running average could drift away from the true mean before returning.

2.2.3 Almost Sure Convergence

On the other hand, this concept is centred around the SLLN, which makes a fundamentally more absolute claim about the behaviour of random variables. Once again, to better understand its mechanics, we must reference the limit equation established in Theorem 2.3:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

This equation fundamentally guarantees statistical stability. Unlike the WLLN limit equation, which evaluates the probability of an error occurring, the SLLN evaluates the probability ($\mathbb{P}$) of the limit itself. It dictates that as the sample size (n) approaches infinity, the sequence of observed sample means ($\bar{X}_n$) will eventually converge to exactly the true theoretical population mean (μ). Crucially, this happens with a probability of exactly 1.

The major theoretical distinction of *almost sure convergence* is that it evaluates the infinite timeline of the data, rather than a single snapshot in time. As the WLLN only evaluates these isolated snapshots, it leaves open the possibility that the running average could occasionally drift away from the mean as n continues to grow, provided these deviations become increasingly rare. *Almost sure convergence* nullifies this drifting behaviour by guaranteeing a point of no return. It dictates that the running average will only drift outside our chosen margin of error a finite number of times. Once the sequence inevitably passes its final deviation, it is permanently trapped within the boundary.

The limit equation in Theorem 2.3 ensures that the probability of the sequence continuously fluctuating or permanently diverging from μ is strictly zero. As we later demonstrate in Figure 2, running an experiment to sufficiently large value of n and plotting the running average $\bar{X}_n$ over time, causes the line to eventually lock onto the value μ and remain there indefinitely [8]. As a result, *almost sure convergence* makes a much more powerful argument about the behaviour of the sequence and is classified as mathematically stronger than *convergence in probability*. If a sequence of random variables satisfies the conditions necessary for *almost sure convergence*, it inherently satisfies the requirements for the WLLN as well, making the SLLN the definitive limit theorem for independent variables.

2.3 Real-world application

In applied statistics, this snapshot seen in the WLLN is highly practical and serves as the foundation for standalone large scale studies. Prime examples of this are demographic surveying and national polling, as noted in Evans’s book of probability and statistics [2, Sec. 5.4.4, Survey Sampling]. If a researcher were to sample 100,000 individuals to determine a population’s average age, then *convergence in probability* would show that the observed average of that specific finite sample reflects the true population average within a tight margin of error.

Another critical application is found in industrial quality control. If a manufacturing plant produces millions of components, then testing each unit independently isn’t practical. Instead, engineers will rely on *convergence in probability*; by testing a sufficiently large random sample, the probability that the sample’s defect rate significantly differs from the factory’s true overall defect rate will be redundant [5]. The WLLN ensures that this single, large sample provides a statistically accurate bases for quality assurance, without having to require an infinite sequence of testing for the result to be accurate.

In applied probability, the continuous long term stabilisation, guaranteed by *almost sure convergence*, serves as a safety-net for institutions that manage continuous risk. A classic example of this use case is casinos, a topic discussed in Grinstead’s ”Introduction to

Probability” [4, Example 6.13, p. 237]. Unlike a researcher who takes a single observation, a casino continuously processes millions of independent wagers over an indefinite timeline. *Almost sure convergence* guarantees that as this sequence of bets extends to an increasingly large number, the casino’s cumulative average return will lock onto the mathematical advantage, serving as the foundation for the classic phrase ”the house always wins”. The SLLN ensures that the running average will not permanently drift into a loss, providing a strong theoretical basis for their business model.

Another application is found in computational mathematics, more specifically machine learning algorithms [9]. When analysts use computer algorithms to estimate highly complex, multidimensional probabilities, they generate a continuous and sequential stream of random variables. *Almost sure convergence* ensures that as the simulation runs and the number of iterations approaches infinity, the running average of these random samples will permanently converge to the exact predicted value. Unlike the WLLN, which might permit the algorithm’s output to occasionally drift away from the true value after a long period, the SLLN assures programmers that their continuous simulations will eventually stabilise with absolute certainty.

2.4 Proof of the Weak Law of Large Numbers

Having explored the conceptual interpretations and real world applications of the limit theorems, it is imperative to formally validate these claims rather than accepting them on intuition alone. Therefore, this section is dedicated to constructing the formal algebraic proof of the WLLN.

2.4.1 Lemma: Chebyshev’s inequality

Before completing the proof of the main theorem, it is necessary to establish the behaviour of Chebyshev’s inequality. The following lemma provides the framework to isolate the deviation.

Lemma 2.4 (Chebyshev’s inequality). *Let X be a random variable with finite expected value $\mathbb{E}[X_i] = \mu$ and finite variance $\text{Var}(X_i) = \sigma^2$. For any real number $\epsilon > 0$, we have the following inequality [7, p. 80]:*

$$(2.4) \quad \mathbb{P}(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

To prove this, we utilise Markov’s Inequality [7, p. 80], which states that for any non-negative random variable Y and any constant $a > 0$:

$$(2.5) \quad \mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}$$

Next, we define a new random variable $Y = (X - \mu)^2$ and set our constant $a = \epsilon^2$. Since the square of any real number is non-negative, $Y \geq 0$. This allows us to apply Markov’s

Inequality directly:

$$(2.6) \quad \mathbb{P}((X - \mu)^2 \geq \epsilon^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\epsilon^2}$$

Finally, we can simplify both sides of the inequality based on two definitions:

1. *The event $(X - \mu)^2 \geq \epsilon^2$ is strictly identical to the event $|X - \mu| \geq \epsilon$.*
2. *By definition, the variance of X is $\sigma^2 = \mathbb{E}[(X - \mu)^2]$.*

Substituting these identities into our inequality yields the final result:

$$(2.7) \quad \mathbb{P}(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

2.4.2 Proof using Chebyshev's Inequality

Having established Chebyshev's Inequality, we can now apply this to formally prove the WLLN. The following proof fundamentally relies on the variance of the sample mean to demonstrate *convergence in probability* [7, sec.6].

Theorem 2.5 (Weak Law of Large Numbers). *Under the conditions of Definition 2.1, as $n \rightarrow \infty$, the sample mean $\bar{X}_n$ converges in probability to the true mean μ [7, sec.6].*

Proof. Under the conditions of Definition 2.1, we can determine the expectation of the sample mean $\bar{X}_n$ by the linearity of expectation [7, sec.5]:

$$(2.8) \quad \mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n}(n\mu) = \mu$$

Because the random variables are independent, the variance of their sum is equal to the sum of their variances [7, sec.6]:

$$(2.9) \quad \text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}$$

Chebyshev's Inequality provides an upper bound on the probability that a random variable deviates from its mean. For any random variable Y with finite mean and variance, and for any fixed $\epsilon > 0$ [7, sec.5]:

$$(2.10) \quad \mathbb{P}(|Y - \mathbb{E}[Y]| \geq \epsilon) \leq \frac{\text{Var}(Y)}{\epsilon^2}$$

We substitute $Y = \bar{X}_n$ into the inequality. Using the properties obtained in the previous steps, we have $\mathbb{E}[Y] = \mu$ and $\text{Var}(Y) = \frac{\sigma^2}{n}$. Therefore:

$$(2.11) \quad \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2/n}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

Finally, we evaluate the limit as the sample size n approaches infinity. Because σ^2 and ϵ are fixed, finite constants:

$$(2.12) \quad \lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{\sigma^2}{n\epsilon^2} = 0$$

Since probability must be non-negative, the limit is exactly 0. Therefore, the sample mean $\bar{X}_n$ converges in probability to the true mean μ . $\square$

2.4.3 Significance of the Result

This proof demonstrates that the stability of the sample mean is a result of the variance decreasing as the sample size increases. Specifically, because $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$, the spread of the sampling distribution collapses around the true mean μ . This justifies the observation that larger datasets yield more reliable estimates, meeting the requirement for the convergence where the likelihood of a "bad" estimate disappears in the limit.

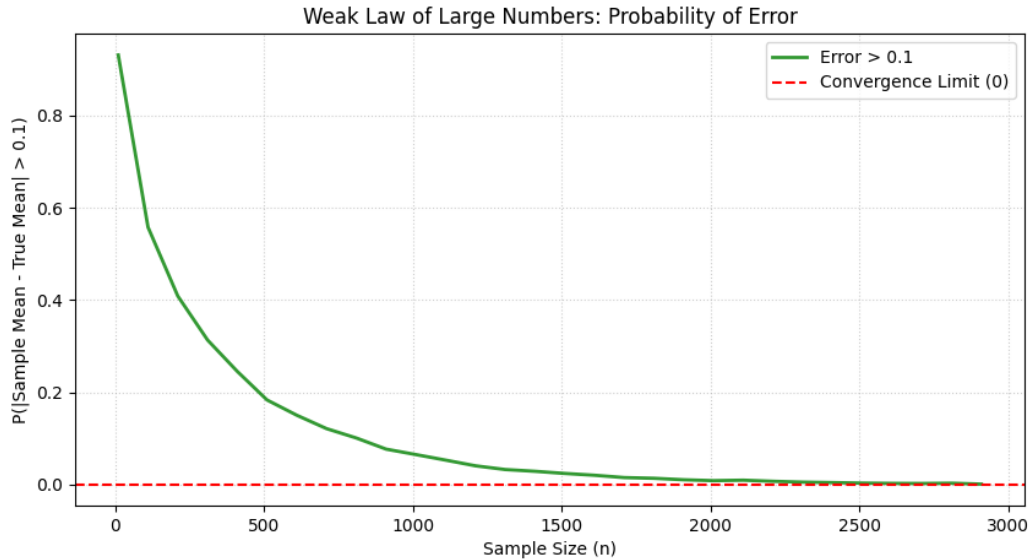


Figure 1: Demonstration of the WLLN. This graph illustrates that as the sample size increases up to a sufficiently large value of n , in this case 3,000. The probability of the sample mean deviating from the true theoretical mean, by a margin greater than 0.1, approaches zero. This visualises the concept of *convergence in probability*, where the likelihood of observing a significant error vanishes entirely for sufficiently large samples.

2.5 Understanding the Strong Law of Large Numbers

To fully appreciate the mathematical rigour of the SLLN, we must investigate the mechanics of the drifting effect, covered in Section 2.2.3, and isolate the specific conditions that prevent it from continuing indefinitely. The fundamental idea that causes *almost sure convergence* is contained within the bounds of probability limits. To explore this concept, we will need to understand two main areas, *The Borel-Cantelli Lemma* and *Kolmogorov's Formulation*, as well as cover the specific notation required for this section [7, sec.4].

To establish *almost sure convergence* for the SLLN, we must demonstrate that the sample mean eventually remains within an arbitrarily small margin of error $\epsilon > 0$. To formalize this continuous timeline, we define an event A_n representing the occurrence where an individual sample mean $\bar{X}_n$ deviates from the true mean μ by at least ϵ :

$$(2.13) \quad A_n = \{|\bar{X}_n - \mu| \geq \epsilon\}$$

To measure the behaviour of these deviations over time, we use infinite unions and intersections. The infinite union, denoted by $\bigcup_{n=N}^{\infty} A_n$, represents the occurrence of at least one deviation from index N onwards. Applying an infinite intersection to these continuous unions, denoted by $\bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$, creates a strict boundary dictating that no matter how large N becomes, a deviation will still eventually occur.

As established by Billingsley [7, pp. 52–53], combining these operations constructs the limit superior of the sequence. This defines an event occurring infinitely often (i.o.):

$$(2.14) \quad \limsup_{n \rightarrow \infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \{A_n \text{ i.o.}\}$$

This notation bridges the conceptual idea of drifting directly to our final proof. For the SLLN to hold, the probability of this entire limit superior must be exactly zero.

2.5.1 The First Borel-Cantelli Lemma

For the sequence of sample means to *converge almost surely*, the probability that deviations occur infinitely often must be strictly zero. Having formalised this requirement as $\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0$, we require a mechanism to prove when this condition is actually met.

The First Borel-Cantelli Lemma establishes that if the infinite sum of the individual snapshot probabilities converges to a finite value, the sequence will *almost surely* stop deviating from the boundary. In Billingsley's formulation [7, Theorem 4.3, p. 59]:

Lemma 2.6 (First Borel-Cantelli Lemma). [7, Theorem 4.3, p. 59] *If the infinite sum of the probabilities of a sequence of events A_n converges, then the probability of their limit superior (the event occurring infinitely often) is exactly zero:*

$$(2.15) \quad \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty \implies \mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) = 0$$

2.5.2 Kolmogorov's Formulation

The First Borel-Cantelli Lemma provides the necessary condition to prevent infinite drifting, but proving that the sum $\sum \mathbb{P}(A_n)$ actually converges for a sample mean requires us to build further on the mathematics we have already seen.

It might seem intuitive to use Chebyshev's inequality similar to how we applied it in the WLLN. However, if we attempt to rely on this bound to satisfy the Borel-Cantelli condition, we encounter a problem with the rate at which it decays. Using Chebyshev's inequality, we find that the probability of a single deviation is bounded by:

$$(2.16) \quad \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

To prove almost sure convergence, the First Borel-Cantelli Lemma strictly requires the infinite sum of these probabilities to be a finite number. If we attempt to sum the Chebyshev bound over all n towards infinity, we can factor out the constants to reveal the underlying sequence:

$$(2.17) \quad \sum_{n=1}^{\infty} \frac{\sigma^2}{n\epsilon^2} = \frac{\sigma^2}{\epsilon^2} \sum_{n=1}^{\infty} \frac{1}{n}$$

The remaining sequence, $\sum \frac{1}{n}$, is the harmonic series. It is a fundamental property of calculus that the harmonic series strictly diverges to infinity. Because this sum does not converge to a finite number, Chebyshev's inequality decays too slowly to satisfy the Borel-Cantelli requirement. This dead end is exactly what necessitates a more powerful bounding method.

We can solve this by using Andrey Kolmogorov's much stricter maximal bound known as Kolmogorov's Inequality [7, sec. 22]. For a sequence of independent variables with finite variance, Kolmogorov's inequality bounds the maximum partial sum rather than just the final sum:

$$(2.18) \quad \mathbb{P} \left(\max_{1 \leq k \leq n} \left| \sum_{i=1}^k (X_i - \mu) \right| \geq \epsilon \right) \leq \frac{\text{Var}(\sum_{i=1}^n X_i)}{\epsilon^2}$$

To apply this to the SLLN, the classical proof evaluates the sequence along an exponentially growing dual subsequence, $n = 2^k$. By applying Kolmogorov's inequality to this subsequence, we obtain a bound strong enough to control the maximum fluctuations between 2^{k-1} and 2^k [7, p. 288]. This crucial step ensures that the sum of the probabilities over the subsequence converges to a finite number [7, p. 289]:

$$(2.19) \quad \sum_{k=1}^{\infty} \mathbb{P} \left(\max_{2^{k-1} < n \leq 2^k} |\bar{X}_n - \mu| \geq \epsilon \right) < \infty$$

We can now directly apply the First Borel-Cantelli Lemma to the subsequence, which then extends to all n . This proves that the sequence of sample means will eventually remain within the margin of error for all future observations [7, p. 59]:

$$(2.20) \quad \mathbb{P} \left(\limsup_{n \rightarrow \infty} A_n \right) = 0$$

This equation states that, for any arbitrarily small margin of error $\epsilon > 0$, deviations will occur only finitely many times.

As this holds true for any $\epsilon > 0$, it implies that the sequence of sample means $\bar{X}_n$ will eventually remain within every possible bound around μ . In the formal language of measure theory, the set of outcomes where the sequence fails to converge to μ has a probability measure of zero [7, p. 268]. Therefore, we can conclude that the sequence of sample means converges to the true population mean *almost surely*, with probability 1:

$$(2.21) \quad \mathbb{P} \left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu \right) = 1$$

□

Thus, we arrive at the formal mathematical conclusion of the SLLN, proving *almost sure convergence*.

2.5.3 Visual Interpretation

Having examined the SLLN from both a conceptual and rigorously mathematical perspective, this section provides a simulated visual example to further illustrate the *almost sure convergence* of the sample mean to the true population mean.

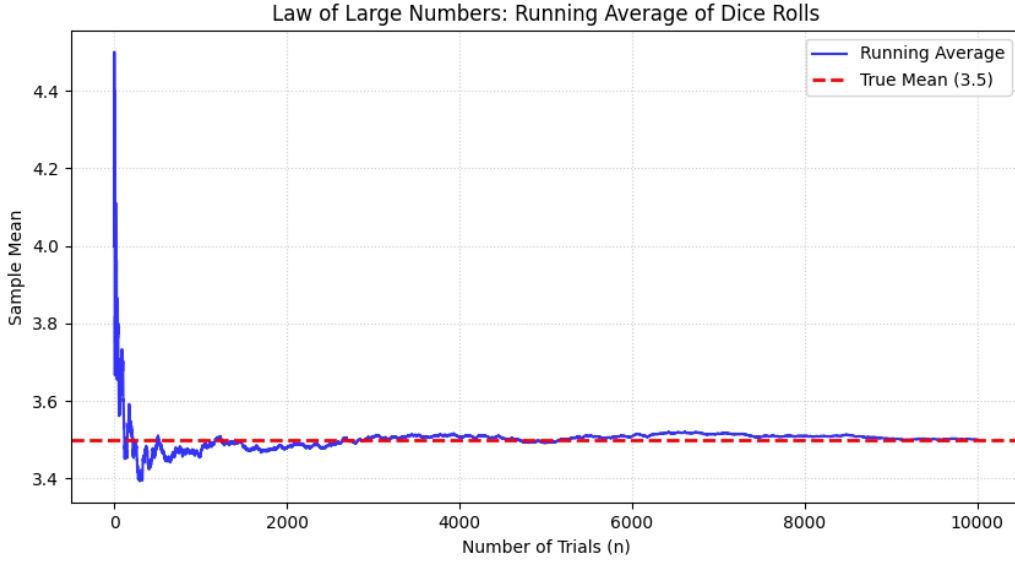


Figure 2: Demonstration of the LLN using dice rolls. This graph illustrates that across 10,000 simulated dice rolls, the running average converges to the true theoretical mean of 3.5. This visualises the idea of *almost sure convergence*, where deviations from the mean vanish entirely.

2.6 The Boundaries of the Theorem: When the Strong Law and Weak Law fail

While proving the success of the limit theorems establishes the foundation of applied probability, introducing counterexamples to expose these vulnerabilities provides a more comprehensive understanding of the theorems and their limits. We can understand these precise conditions under which these theorems fail by introducing probability distributions that violate fundamental prerequisites, such as the existence of a finite expected value. This then allows us to observe the exact thresholds where these theorems collapse. To systematically explore these limits, we can look at the breakdown of convergence in probability under the WLLN, followed by the failure of *almost sure convergence* from the SLLN.

2.6.1 Counter Model: The Cauchy Distribution

If we wish to test the boundaries of the limit theorems, we must introduce a probability model that intentionally violates their prerequisites. We can achieve this by introducing the Cauchy distribution, which serves as the quintessential counterexample in probability theory, and works because its unique shape prevents the calculation of a finite expected value or variance.

Definition 2.7 (The Cauchy Distribution). [7, sec. 20] Let X be a continuous random

variable. X is said to follow a Cauchy distribution if its Probability Density Function (PDF) is given by:

$$(2.22) \quad f(x; x_0, \gamma) = \frac{1}{\pi\gamma \left[1 + \left(\frac{x-x_0}{\gamma} \right)^2 \right]}$$

where $x_0 \in \mathbb{R}$ represents the location parameter (the median) and $\gamma > 0$ represents the scale parameter.

The defining characteristic of the Cauchy distribution is its heavy-tails. The probability density decays slowly enough that as $x \rightarrow \pm\infty$, the integral required to calculate the first moment (the expected value) strictly diverges. Consequently, the distribution has no true population mean (μ), and as demonstrated in the subsequent sections, this absence fundamentally breaks convergence in the limit theorems.

2.6.2 Collapse of the Weak Law of Large Numbers

The WLLN breaks down in the absence of a finite expected value. To visualise this failure, we can apply the standard Cauchy distribution to our probability model. One unique property of the standard Cauchy distribution is that the sample mean of n independent Cauchy variables follows the exact same standard Cauchy distribution as a single observation. As a result of this property, the distribution of the sample mean does not fix around a central value as n increases significantly.

When we simulate the sequence of sample means and define a margin of error ($\epsilon = 1.0$) around the true mean ($\mu = 0$), the contrast between the initial volatility of the sequence and its eventual stabilisation is shown. Unlike the converging model where the probability of error rapidly converges to zero, the Cauchy model's error probability remains constant at exactly 50%. The visual simulation proves that no matter how infinitely large the sample size becomes, the uncertainty will never shrink, cleanly demonstrating the collapse of the WLLN's fundamental principle of *convergence in probability*.

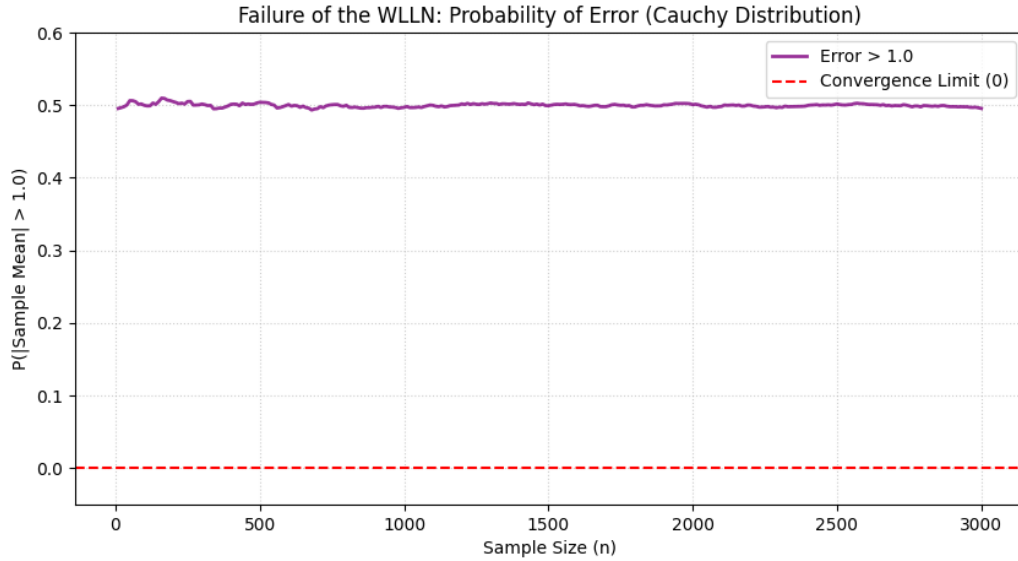


Figure 3: Failure of the WLLN using a Cauchy distribution. This graph tracks the probability that the sample mean of a standard Cauchy distribution deviates from its true mean ($\mu = 0$) by a margin of error ($\epsilon = 1.0$). Calculated across 10,000 independent experiments, the visualisation confirms that the probability of error remains constant at approximately 50%. Due to the fact the distribution lacks a finite expected value, the error rate does not decay to zero as the sample size increases, therefore violating the core principle of *convergence in probability*.

2.6.3 Collapse of the Strong Law of Large Numbers

The LLN is highly robust, and its guarantee is strictly conditional, but it relies on the underlying sequence of random variables possessing a finite expected value (μ). When we introduce a distribution with heavy-tails, this finite moment ceases to exist, and the fundamental idea of the theorem breaks down. To demonstrate this collapse, we make use of the standard Cauchy distribution.

The Cauchy distribution produces extreme outliers in such a frequent way that the integral used to calculate its expected value will diverge, meaning it has no defined population mean. As there is no true mean to keep the data consistent, the sample mean never stabilises. We can see this from the demonstration in the following simulation, where a single extreme outlier can occur at any time and cause the running average to deviate drastically off course, no matter how large the sample size (n) becomes. In the real world, we see that if a phenomenon follows a similar distribution, taking an average of past data would be redundant.

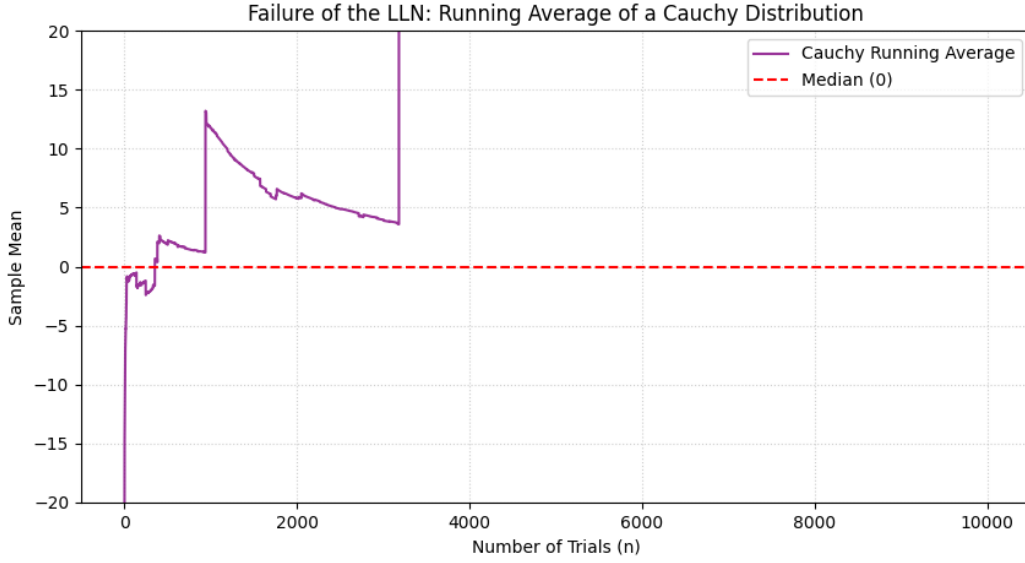


Figure 4: Failure of the LLN using a Cauchy distribution. This graph illustrates the cumulative running average of 10,000 samples taken from a standard Cauchy distribution. Due to this distribution not possessing a finite expected value (a defined true mean), the sequence will never stabilise. As visualised by the extreme sudden spiking in the graph which drastically causes the average to drift, heavy-tailed outliers consistently disrupt the running average timeline, exemplifying that the fundamental conditions required for convergence are not met.

3 The Central Limit Theorem

3.1 Understanding The Central Limit Theorem

In this section, we break down what the Central Limit Theorem (CLT) tells us in terms of probability by defining it along with the key concepts used in its formal proof. We can give an interpretation of what the CLT is with a formal numerical definition, followed by looking at the idea of Convergence in Distribution, Moment Generating Functions (MGF) and Taylor series.

3.1.1 Defining the Central Limit Theorem

The CLT is a foundational theorem in probability theory that expresses the characteristics of sample averages. It suggests that as the sample size of independent identically distributed trials increases, the sampling distribution of the sample mean approximates a normal distribution, irrespective of the original population distribution [3]. In effect, this

guarantees the ability to apply standard statistical methods to calculate probabilities and error margins for complex and real-world data (provided that the sample size is sufficient).

Theorem 3.1 (Central Limit Theorem). *Consider a sequence of independent and identically distributed (i.i.d.) random variables $X_1, X_2, \dots, X_n$. Assume each variable possesses a finite expected value $\mathbb{E}[X_i] = \mu$ and a strictly positive, finite variance $\text{Var}(X_i) = \sigma^2$. As the sample size $n \rightarrow \infty$, the standardised sum converges in distribution to the standard normal distribution $\mathcal{N}(0, 1)$ [3, p. 194]:*

$$(3.1) \quad \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

While the SLLN dictates exactly where the sample mean will eventually converge, it does not describe the behaviour or shape of the fluctuations along the way. The CLT answers this by introducing *convergence in distribution*.

This theorem standardises the sample mean, which reveals the shape of these fluctuations. By subtracting the true population mean μ and dividing by the standard error $\sigma/\sqrt{n}$, the sequence of errors is scaled, centering the expected value precisely at 0 and normalising the variance to 1.

Crucially, this formulation provides the exact method required to transition from the absolute certainty of the SLLN to the CLT, which gives us the framework we need to deal with finite data, allowing us to draw Confidence Intervals around our estimates.

3.1.2 Convergence in Distribution

Formally, Convergence in Distribution is defined as;

Definition 3.2 (Convergence in Distribution). Let $X_1, X_2, \dots$ be a sequence of random variables with cumulative distribution functions (CDFs) $F_1(x), F_2(x), \dots$ respectively. Let X be our target random variable with CDF $F(x)$ [7, sec. 25]. We say that the sequence X_n converges in distribution to X , denoted as $X_n \xrightarrow{d} X$, if:

$$(3.2) \quad \lim_{n \rightarrow \infty} F_n(x) = F(x)$$

for all real values of x at which the target function $F(x)$ is continuous.

This definition demonstrates that with Convergence in Distribution we aren't necessarily concerned with numerical values themselves but rather the overall shape of its probabilities. We can see from the last line that as the Limit of n tends to infinity, the curve of the sample's cumulative distribution function, $F_n(x)$, perfectly overlaps with the curve of the target distribution, $F(x)$. In reference to the Central Limit Theorem, this utilises convergence in distribution to prove that the cumulative probabilities of a standardized sample mean will eventually match the CDF of a standard normal distribution, $\mathcal{N}(0, 1)$.

3.1.3 Moment Generating Functions

MGF are defined as;

Definition 3.3. Let X be a random variable. The MGF of X , denoted as $M_X(t)$, is defined as the expected value of e^{tX} [7, sec. 22]:

$$(3.3) \quad M_X(t) = \mathbb{E}[e^{tX}]$$

Assuming that the expectation exists for all t in some open interval $(-h, h)$ around 0.

In principle, the CLT is dependent on what happens when you add up a large number of independent random variables to find a sample mean $(X_1 + X_2 + \dots + X_n)$. This definition of MGF allows us to essentially avoid converting the sum into a product of their individual functions.

More precisely, if X and Y are independent random variables, the MGF of their sum is simply the product of their individual MGF, given by:

$$(3.4) \quad M_{X+Y}(t) = M_X(t)M_Y(t)$$

Evaluating the limit as $n \rightarrow \infty$ gives us this underlying normal distribution we are looking for.

3.1.4 Taylor Series

For proving the CLT we need to look at the specific properties of the data, in particular the mean and the variance from the MGF. This is achieved by using a Taylor series expansion.

A Taylor series allows us to approximate complex functions as an infinite sum of polynomial terms, if we expand the exponential function e^{tX} seen from our definition of our MGF and take the expected value, we see that the MGF can be written as an infinite series of its moments [7, sec. 22]:

$$(3.5) \quad M_X(t) = 1 + t\mathbb{E}[X] + \frac{t^2}{2!}\mathbb{E}[X^2] + \frac{t^3}{3!}\mathbb{E}[X^3] + \dots$$

As we are looking at specifically the mean and variance, we only need to focus on the first and second moments of our expansion. We will denote the remaining terms we have truncated from the equation as $o(t^2)$. This implies our MGF can be written as;

$$(3.6) \quad M_X(t) = 1 + t\mathbb{E}[X] + \frac{t^2}{2}\mathbb{E}[X^2] + o(t^2)$$

In reference to the CLT, this truncated series is what removes the original shape of the data and provides us with our shape similar to that of the normal distribution. Before calculating the final limit, we need to standardise our random variables. If we let Z

be a standardised random variable, we know that $(\mathbb{E}[Z] = 0)$ and its variance will be $(\mathbb{E}[Z^2] = 1)$. Substituting in these values into our Taylor series expansion, we achieve;

$$(3.7) \quad M_Z(t) = 1 + t(0) + \frac{t^2}{2}(1) + o(t^2)$$

$$(3.8) \quad M_Z(t) = 1 + \frac{t^2}{2} + o(t^2).$$

By eliminating the linear term and fixing the quadratic term at 1, we can isolate the variance. This ensures that as the sample size $n \rightarrow \infty$, the complex shape of the parameters that $o(t^2)$ contained, when truncating the original series, will vanish. This is an essential idea that can be later applied to the CLT proof.

3.2 Proof of the Central Limit Theorem

The following theorem is based off of Billingsley's book on "Probability and Measure" [7, sec. 27].

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with mean μ and finite variance σ^2 .

Initially, we must standardise our random variables. Therefore, let us define a new standardised random variable, Z_i , for each observation:

$$(3.9) \quad Z_i = \frac{X_i - \mu}{\sigma}$$

Because Z_i is standardised, its expected value is $\mathbb{E}[Z_i] = 0$ and its variance is $\text{Var}(Z_i) = \mathbb{E}[Z_i^2] = 1$.

We then need to analyse the behaviour of the sample mean, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Once again we need to standardise the sample mean itself to observe its convergence to a standard normal distribution, $\mathcal{N}(0, 1)$. Let Z_n^* denote the standardised sample mean:

$$(3.10) \quad Z_n^* = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i$$

With this, we can apply our MGF to examine the distribution of Z_n^* . As established, if variables are independent, the MGF of their sum is the product of their individual MGF. Therefore, the MGF of the sum $\sum_{i=1}^n Z_i$ is $(M_Z(t))^n$. However, because our standardised sample mean Z_n^* is scaled by $\frac{1}{\sqrt{n}}$, we must scale the parameter t equally:

$$(3.11) \quad M_{Z_n^*}(t) = \left(M_Z \left(\frac{t}{\sqrt{n}} \right) \right)^n$$

Substituting in our Taylor series expansion for the MGF of a standardised variable, we truncate the series to isolate the variance which leaves us with $M_Z(t) = 1 + \frac{t^2}{2} + o(t^2)$. Substituting $\frac{t}{\sqrt{n}}$ into this expansion give us:

$$(3.12) \quad M_Z\left(\frac{t}{\sqrt{n}}\right) = 1 + \frac{1}{2}\left(\frac{t}{\sqrt{n}}\right)^2 + o\left(\left(\frac{t}{\sqrt{n}}\right)^2\right) = 1 + \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)$$

Plugging this back into the MGF equation for our standardised sample mean, we get:

$$(3.13) \quad M_{Z_n^*}(t) = \left(1 + \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n$$

To determine the final shape of the distribution, we evaluate the limit as the sample size n approaches infinity. Referring back to the definition for the exponential function: $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$. Applying this limit, the complex parameter shapes contained within the remainder term $o(t^2/n)$ vanish as $n \rightarrow \infty$:

$$(3.14) \quad \lim_{n \rightarrow \infty} M_{Z_n^*}(t) = e^{t^2/2}$$

The resulting function, $e^{t^2/2}$, is precisely the MGF of the standard normal distribution, $\mathcal{N}(0, 1)$. Because the limit of the MGF matches the MGF of a standard normal distribution, Convergence in Distribution dictates that the cumulative probabilities of the standardised sample mean perfectly overlap the standard normal curve as $n \rightarrow \infty$. This successfully proves that the sampling distribution of the sample mean approximates a normal distribution, regardless of the original population distribution. $\square$

3.2.1 Significance of the Result

The proof of the CLT reveals a remarkable phenomena in probability theory; the emergence of the normal distribution. Unlike the LLN, which only guarantees that a sample mean will eventually converge to a single value, the CLT dictates the exact shape of the variance around that value.

The main significance of this result lies in the nature of the distribution itself. Provided the underlying population possesses a finite mean and variance, we can see that the original shape of the raw data, irrespective of if it's perfectly flat (similar to a uniform distribution), heavily skewed, or completely unknown, becomes entirely irrelevant as the sample size grows. As proven by the MGF expansion, the process in which we standardise all the variables mathematically erases the original distribution's complex parameters in the limit, which in affect forces the distribution of the sample means into a symmetrical and predictable bell curve ($\mathcal{N}(0, 1)$).

In practice, we see this theorem in statistics is the foundational justification for almost all conclusive modelling. Since the sample mean is guaranteed to be normally distributed

for large samples, statisticians can reliably calculate Confidence Intervals, perform hypothesis testing, and assess margins of error without ever needing to have known the true shape of the population data. Together with the Python simulation, we can perfectly encapsulate this significance. By extracting thousands of sample means from a strictly non-normal uniform distribution, the empirical data constructs a perfect theoretical normal curve, providing us with a more concrete and visual proof of the theorem’s capabilities.

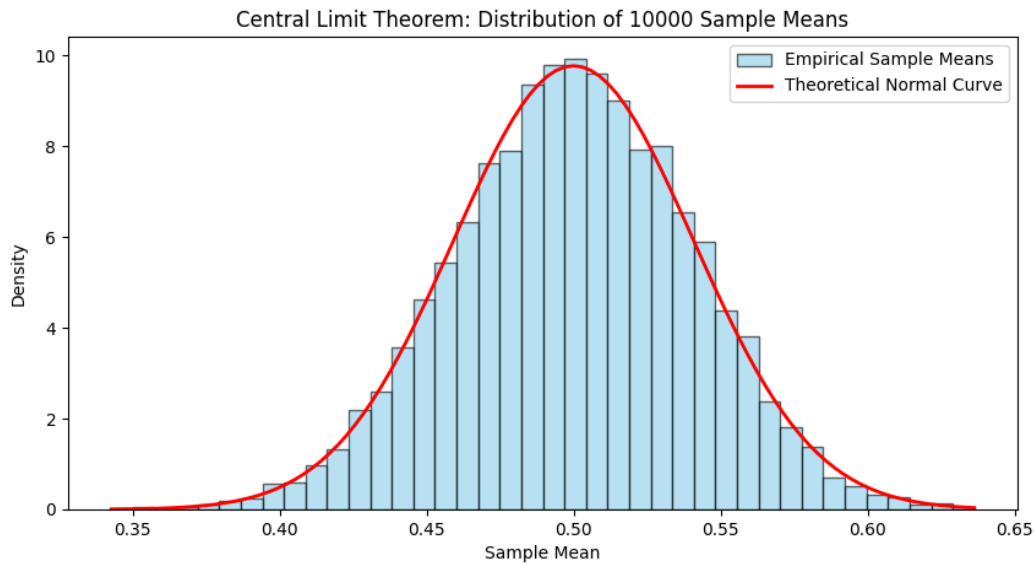


Figure 5: The histogram above shows the distribution of 10,000 empirical sample means (each of size $n = 50$) drawn from a non-normal uniform distribution. We can see that distribution of the sample means almost perfectly mirrors the theoretical normal bell curve (red line).

3.3 The Boundaries of the Theorem: When the CLT Fails

While the CLT is often regarded as one of the most robust tools in applied statistics, its guarantee is highly volatile when implemented outside its ideal parameters. The theorem relies on a strict set of the following prerequisites: the underlying population must possess a finite variance, the standardisation must scale exactly by $\sqrt{n}$, and the random variables must be sufficiently independent. The objective of this section is to investigate the exact mathematical boundaries of the CLT. To achieve this, we can systematically dismantle the theorem by intentionally violating these prerequisites. This methodology allows us to isolate the specific cause for when the theorem collapses. Through a series of Python based simulations, we can explore two distinct failure states: the absence of finite variance, and the violation of independence through infinite memory. To facilitate these investigations we utilise specific distributions, namely the Pareto and Cauchy distributions, which serve as theoretical counter models to the standard Gaussian assumption. By formally defining

these distributions and their associated moment conditions, we can isolate the exact point at which the CLT prerequisites are no longer upheld.

3.3.1 The Cauchy Distribution

As defined in Definition 2.7, the Cauchy distribution serves as the failure state for statistical convergence that is utilised in this section as well. While we previously observed how its lack of an expected value causes the LLN to collapse, its lack of a finite variance is equally harmful to the CLT. For the same reason that the variance evaluates strictly to infinity, standardising the sample mean by dividing by the standard deviation becomes mathematically impossible. As a result, the sample mean never settles into a normal approximation and remains as random and dispersed as the individual observations, regardless of how large the sample size becomes.

3.3.2 The Pareto Distribution

The Pareto distribution provides a controlled environment to look at the behaviour of heavy-tailed phenomena. This is defined strictly on the interval $[x_m, \infty)$, as its probability density function (PDF) is given by [3, chapter 4]:

$$(3.15) \quad f(x; \alpha, x_m) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}} \quad \text{for } x \geq x_m$$

where $x_m > 0$ denotes the scale parameter and $\alpha > 0$ denotes the shape parameter.

The importance of the Pareto distribution in this section lies in the existence of its moments, which are strictly dictated by the value of α . Specifically, when $1 < \alpha \leq 2$, the first moment (the expected value) is finite, ensuring that the LLN remains satisfied. However, the second central moment (the variance) evaluates to infinity. This property allows us to intentionally isolate and violate the CLT finite variance prerequisite while leaving the mean intact.

3.3.3 The Infinite Variance Assumption (The Pareto Model)

Similar to how the Law of Large Numbers requires a finite mean, the Central Limit Theorem strictly requires the underlying population to have a finite variance (σ^2). If the variance is infinite, the theorem is guaranteed to collapse. To identify this failure, we can sample from a Pareto distribution with a shape parameter of $\alpha = 1.5$. This distribution provides a perfect theoretical boundary, as $\alpha > 1$, the distribution possesses a finite expected value (meaning the Law of Large Numbers still holds). However, because $\alpha \leq 2$, its variance is infinite. The mathematical proof of the Central Limit Theorem relies on standardising the sample mean by dividing it by the population standard deviation. Without a finite variance, this standardisation wouldn't be possible, and the distribution of the sample means

can't converge to the standard normal distribution $\mathcal{N}(0, 1)$. Evident from what is seen in the simulation, instead of a symmetrical bell curve, the resulting distribution is severely right skewed and heavily tailed. In practise, if a statistician simply assumes the Central Limit Theorem applies to such heavy tailed data, they will drastically underestimate the probability of extreme events.

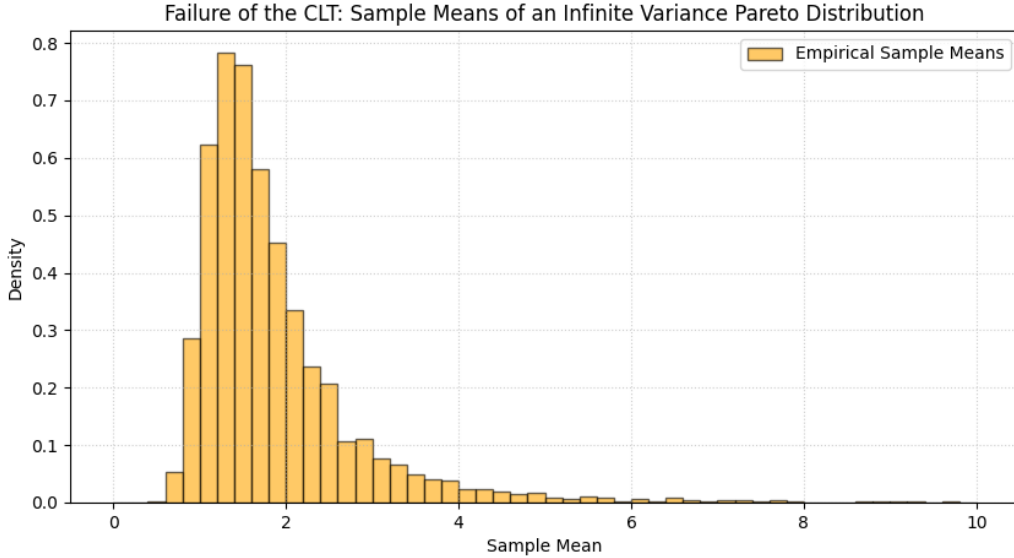


Figure 6: Failure of the Central Limit Theorem using a Pareto distribution. This histogram displays the distribution of 10,000 sample means (each of size $n = 50$) taken from a Pareto distribution with a shape parameter of $\alpha = 1.5$. Because this distribution possesses an infinite variance, the sample means fail to converge into an expected symmetrical standard normal bell curve. Instead, the distribution remains heavily right-skewed, visually proving that the theorem's assumption of finite variance is a strict requirement for normal approximation.

3.3.4 The Independence Axiom - Strong Mixing Collapse

Having established the foundational prerequisites for the CLT, we can now investigate its vulnerability to violations of the independence axiom. While the CLT can adapt to certain time series that contain minor dependencies (provided the sequence satisfies Strong Mixing (or α -mixing) conditions where dependence decays to zero over time), it fundamentally collapses when this decay does not occur.

To isolate this failure mathematically, we restrict our testing environment to a standard uniform distribution, $\mathcal{U}(0, 1)$. This ensures that both the expected value ($\mu = 0.5$) and the variance ($\sigma^2 = 1/12$) are strictly finite, intentionally avoiding the heavy-tailed infinities that caused our previous failures. Instead, we break the CLT by selecting random variables

that simply do not work within its framework, and introducing permanent infinite memory into the sequence.

In this simulation, every variable within a single sample is perfectly correlated to a shared underlying factor. Because this correlation never decays, the random variables fail to act independently and cannot mix. To demonstrate the collapse, we maintain the exact CLT scaling factor ($\sqrt{n}$) by calculating the standardised Z -scores for our sample [7, sec. 27] :

$$(3.16) \quad Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

As visualised in Figure 7, this infinite dependence breaks the limit theorem. Rather than the standardisation process scaling the aggregated data into a standard normal bell curve with a variance of 1, the $\sqrt{n}$ scaling factor amplifies the perfectly correlated errors. Consequently, the standardised distribution retains the exact flat, rectangular shape of the original uniform distribution, but is stretched massively across the x -axis. This proves that without independent variables, the variance of the sum fails to normalise, rendering the CLT ineffective.

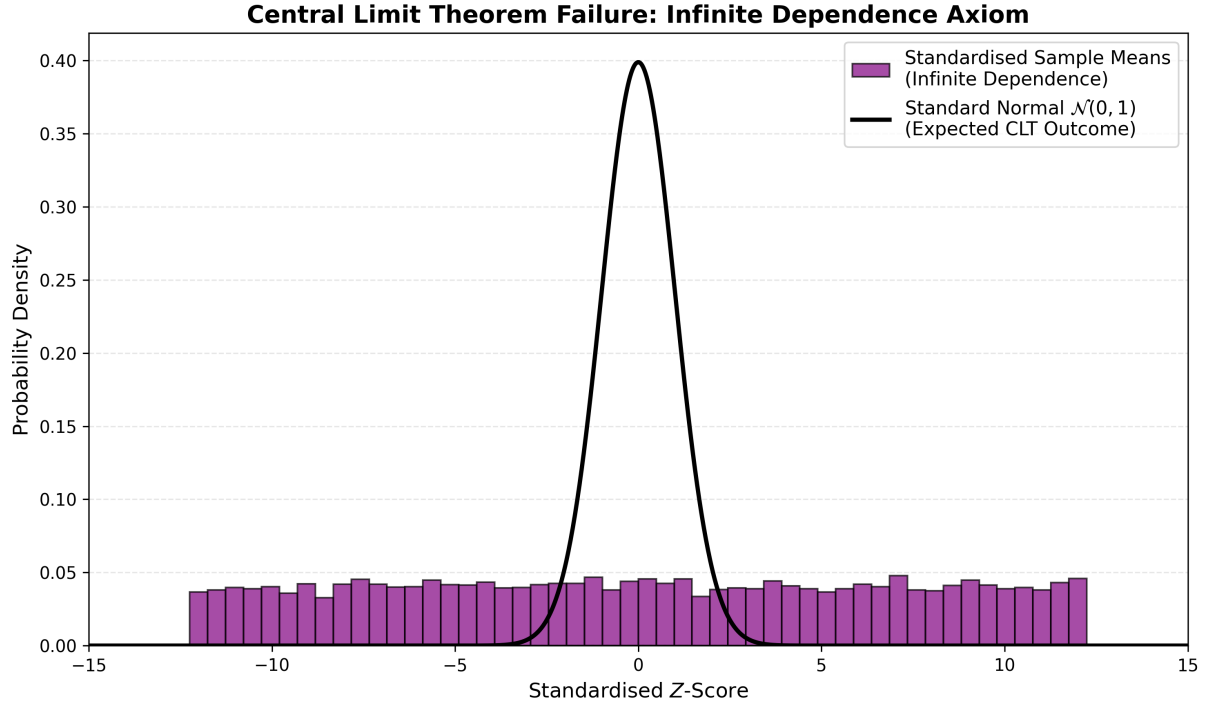


Figure 7: Failure of the CLT under infinite dependence. This histogram displays the standardised Z scores of 10,000 sample means (each of size $n = 50$) drawn from a standard uniform distribution. By intentionally violating the strong mixing requirement and introducing infinite memory, the variables fail to act independently. Even when applying the exact CLT standardisation formula ($\sqrt{n}$ scaling), the data fails to approximate the expected normal curve. Instead, it perfectly mirrors the flat shape of the parent uniform distribution.

4 Conclusion

The aim of this paper was to investigate the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT) through two distinct approaches: rigorous mathematical proofs and graphical visualisation. By exploring these fundamental theorems from both these theoretical perspectives, this project successfully demonstrates how simple and predictable data can emerge from intrinsic randomness.

Through these mathematical proofs, we were able to establish the core mechanisms of these theorems and demonstrate that the LLN guarantees the convergence of a sample mean to its population mean. Furthermore, by utilising Convergence in Distribution, Moment Generating Functions, and Taylor series expansions, we successfully proved that the CLT dictates the shape of this convergence, showing that the sampling distribution approximates a normal distribution, gives us our bell curve regardless of the original population's shape.

The second approach converted these concepts into something more tangible. The simulations successfully showed the theoretical proofs in action. By modelling the running average of dice rolls, we were able to confirm the stabilising effect of the Strong Law of Large Numbers over time. Similarly, by repeatedly sampling from a uniform distribution, the emergence of the bell curve allowed for a practical verification of the CLT.

Crucially, the investigation into the Cauchy distribution as a counter example highlighted the strict mathematical boundaries of these theorems. By creating a scenario where the running average fails to settle, it reinforces the necessity of the fundamental conditions contained within both theorems and that the sequence of random variables must possess a finite expected value and a finite variance.

Ultimately, this dual approach was able to bridge the gap between theoretical probability and applied statistics. The mathematical proofs provide the absolute certainty that these limit theorems hold true, whilst the simulations offer the practical intuition necessary to apply them to real world data analysis. Together, they consolidate the LLN and the CLT as crucial foundations of modern statistics.

5 Final Reflections

The process of completing this project has been both enjoyable and challenging, as I have gone from only ever learning about mathematical theorems to actively interrogating them. Initially, my goal was simply to prove the LLN and the CLT. However, as my research progressed, my initial project plan evolved significantly. Prompted by my increased research into specific topics, as well as discussions with my supervisor, I shifted my focus from merely demonstrating where these theorems succeed, to systematically investigating their boundaries and finding out what axioms I can experiment with to explore their behaviour when they collapse.

This pivot required me to engage with concepts well outside of what was taught in my modules. Understanding the exact boundaries of the LLN and CLT forced me to study heavy-tailed mathematics, specifically the Cauchy and Pareto distributions. Initially, grasping how a distribution could exist without a finite expected value or variance, and understanding how that could be used in my project was a significant hurdle. Furthermore, formally proving the CLT required me to independently learn and apply MGF and Taylor series expansions. Mastering these techniques was intrinsically challenging, but it provided a deeply satisfying realisation of how standardisation erases a distribution's original parameters in the limit.

Alongside the theoretical mathematics, a major skill I developed was programming using Python. Before this project, my programming experience was very elementary, having only practised Python when necessary such as for module requirements. Translating complex algebraic prerequisites, such as the finite variance axiom, into functional code to generate graphical proofs was a steep learning curve. This computational aspect is also where I encountered my most instructive mistake, where when attempting to simulate the

breakdown of standardisation for the CLT, I initially struggled to understand why my empirical sample wasn't deforming as expected. In an attempt to cover all prerequisite fail cases, I ended up going down a rabbit hole of changing the standard scaling for the CLT, which resulted in me producing uninteresting results. It had simply boiled down to producing a result which either tended towards 0 if I scaled by $< n^{\frac{1}{2}}$, or a result that tended toward infinity if I scaled $> n^{\frac{1}{2}}$. This caused significant time pressure as I then needed to correctly amend this error and properly demonstrate the fail case for when the variance was finite and we have the standard scale factor. Instead, altering the random variables themselves caused a disruption to the alpha standard distribution which yielded the result I was actually trying to achieve. This taught me to revisit the core definitions of the theorems, as I had now learned that coding mathematics requires an absolute, rigorous understanding of the underlying equations and any minor mistake would cause the entire process to fail.

Another major challenge I faced during this project was understanding how to properly structure each section. After reviewing my work across multiple meetings with my supervisor and incorporating their feedback, I completely reorganised the flow of my arguments. This taught me the importance of creating a logical progression in mathematical writing, rather than just providing the content, as ensuring that each lemma, proof, and simulation is built naturally upon the last, producing a much more polished and rigorous piece of work.

Ultimately, navigating these difficulties taught me the value of independent research. The challenge of finding specialised sources, revisiting complex proofs, and debugging Python scripts required a high level of resilience. Looking back at my initial project plan, I am proud of how much this research has evolved, as transforming a preliminary concept into a rigorous mathematical dissertation has been deeply rewarding, particularly within a field of statistics I find so interesting.

In reflection of this project, my appreciation for applied statistics and probability has grown, although challenging at times, venturing beyond the standard curriculum reminded me exactly why I chose to study mathematics in the first place. Moving forward, I feel significantly more confident in my ability to tackle unseen problems, program simulations, and rigorously challenge the mathematical assumptions we so often accept at face value.

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